

AI/ML Capstone Project

D1 – Project Proposal (5% of final mark)

Due: Week 4 (softcopy on SLATE, hardcopy to your professor in class)

Name 1: _____

Student Number 1: _____

Name 2: _____

Student Number 2: _____

Name 3: _____

Student Number 3: _____

Name 4: _____

Student Number 4: _____

Name 5: _____

Student Number 5: _____

Instructions:

- Please work on the following deliverable with your group.
- Prepare your deliverable report as discussed in class.
- Type your report using word processing software (preferably using LaTeX, otherwise MS Word is fine).
- Upload your report in PDF format on SLATE. Team leader should upload this deliverable on SLATE as a single PDF file.
- Present a printout (stappled) of your deliverable to your professor in class in **Week 4**.

Q1: Which sector driving modern economies are you considering for your capstone project (i.e., **agriculture, education, energy, entertainment, finance, government, health care, insurance, manufacturing, NGO**). Select the sector that is mostly relevant to your capstone. Briefly explain your sector selection and what experience you and your group have in this area. Answer this question in one paragraph. **[0.5 mark]**

Q2: Identify a **real-world problem** that you and your group are interested to explore. Explain the identified “problem” in as much detail that you can. Provide citable data or context to prove that this problem exists and that it matters. Do not forget the famous saying that: “*a problem well stated, is a problem half-solved*”. Answer this question in two to three paragraphs. [1 mark]

Q3: Identify the **prospective clients** and **stakeholders** that the real-world problem that you explained above is targeted for. What are their specific requirements? How are they currently solving this challenge without your AI/ML solution? Support your ideas with reliable market research data. Answer this question in one paragraph. [0.5 mark]

Q4: Explicitly justify **why an AI/ML solution is needed** here rather than applying standard heuristics or traditional software engineering alternatives? What are the benefits of using AI/ML to tackle this real-world problem? Answer this question in one paragraph. [0.5 mark]

AI/ML solutions is needed to collect ,process and analyze real time data from camera feeds , alert message or text data from government about road work/constructions with weather data on the go. Traditional software engineering alternatives relies on rule based logics which cannot process large data and unstructured data successfully. AI/ML solution provides the scalability option ,where several types of data can be analyzed simultaneously using natural language processing for text , neural network for find optimal solution based on number of factors and then logistics regression for the final decision.

Q5: Explain **how AI/ML can be integrated** into the real-world problem you identified above. Support your ideas from existing literature and/or industry benchmarks. Answer this question in two to three paragraphs. [1 mark]

To make this work AI/ML solution applied using , the system uses natural language processing to read emergency text alerts [3] and a convolutional neural network to visually spot surface hazards [6]. These insights are then combined with real-time weather data using a logistic regression model to generate a segment-specific safety score, relying on proven multi-modal fusion strategies [7].

By focusing on catching high-risk conditions rather than just general accuracy [5], the tool proactively warns drivers about severe dangers before the first collision even occurs.

Q6: The AI/ML capstone project expects groups to work with **datasets**. In general, groups can **train, validate, and test an AI/ML model** of their choice. Alternatively, groups may consider **customizing an SLM or LLM with specific or proprietary data** using techniques like fine-tuning or RAG. Provide some insights on where will your data come from? Is the data available in open-source, do you need to scrape, generate it on your own, or synthetically simulate it? Provide explanation on licensing, data privacy, and the volume sufficiency needed for training, model adaptation or context injection. Also, is the considered dataset properly labelled or not and what implications will this have on your overall work. Answer this question in three to four paragraphs. [1 mark]

Q7: An AI/ML technology stack is the generally composed of the **infrastructure layer** (i.e., HW/SW), **data layer** (i.e., injection/cleaning/storage), **framework and model layer** (i.e., TensorFlow/PyTorch, LLM/SLM, fine-tuning/RAG), **MLOps layer** (e.g., Hugging Face, Docker), and the **application layer** (e.g., APIs, UI/Streamlit). Identify a preliminary technology stack suited to tackle the AI/ML challenge of your capstone project. Answer this question by providing a brief sentence for each AI/ML technology stack. **[0.5 mark]**

References (Include 5-7 peer-reviewed sources to back your AI/ML capstone project ideas cited above with your answers).

[1] ?

[2] ?

[3] ?

[4] ?

[5] ?

[6] ?

[7] ?

Good Luck!